MIT | Academy of Engineering
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

KOMPALWAR GANGOTRI DATTA

B.Tech. - Electronics and Telecommunication Engineering

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PROFESSIONAL SUMMARY

Final-year B.Tech. Electronics and Telecommunication Engineering student specializing in AI/ML, with hands-on experience designing and developing intelligent applications across Computer Vision, Generative AI, Retrieval-Augmented Generation (RAG), and Voice AI. Strong technical proficiency in Python, PyTorch, TensorFlow, OpenCV, YOLOv8, FastAPI, LLMs, embeddings, vector databases, and semantic search. Knowledge of LangChain and LangGraph for building RAG pipelines, tool/function calling, agentic workflows, state management, and multi-step AI execution. Experienced in implementing practical AI solutions involving real-time object detection, document intelligence, identity verification, multilingual AI tutoring, speech-to-text, text-to-speech, and AI backend integration. Interested in developing scalable, reliable, and production-oriented AI solutions in Generative AI, Computer Vision, Agentic AI, and Voice AI.

KEY EXPERTISE

Artificial Intelligence & Machine Learning | Deep Learning | Generative AI | Retrieval-Augmented Generation (RAG) | Computer Vision | Python | TensorFlow | PyTorch | OpenCV | YOLOv8

EDUCATION

MIT Academy of Engineering, Pune 2023 - 2027
B.Tech. - Electronics and Telecommunication Engineering | CGPA: **7.90** / **10**

INDIRA INSTITUTE OF TECHNOLOGY(POLY), Nanded 2024
Diploma | Diploma - Computer Engineering | MSBTE | Percentage: **87.49** / **100**

N.E.S.SCIENCE COLLEGE SHRI NAGAR, Nanded 2021
12th | MSBSHSE | Percentage: **94.17** / **100**

Pratibha Niketan High School, Nanded 2019
10th | MSBSHSE | Percentage: **93.80** / **100**

INTERNSHIPS

SmartBridge (SkillWallet) | Remote | IT / Computers - Software 16 Jun, 2025 - 05 Aug, 2025
Artificial Intelligence and Machine Learning Intern

Key Skills: Python | Machine Learning

Completed an experiential internship in Artificial Intelligence and Machine Learning at SmartBridge (SkillWallet), gaining hands-on experience in Python-based machine learning workflows. Performed data preprocessing, feature engineering, model development, and performance evaluation. Applied machine learning algorithms and predictive analytics techniques to solve real-world problems, strengthening practical skills in AI/ML model development and implementation.

PROJECTS

GramNet AI: Offline Multilingual RAG-based AI Tutor

Key Skills:

RAG | LLMs | TinyLLaMA | Semantic Search | FAISS | SentenceTransformers | Python | SpeechRecognition/Google STT | Edge TTS | pyttsx3 | LangChain

Project Link: https://github.com/kompalwargangotri/gramnet-ai-offline-multilingual-rag-tutor

Developed an AI-powered multilingual tutoring system using Retrieval-Augmented Generation (RAG) and TinyLLaMA to deliver context-aware educational responses. Implemented a semantic retrieval pipeline using text chunking, SentenceTransformers embeddings, and FAISS for efficient knowledge retrieval and context selection. Integrated a locally hosted TinyLLaMA model through llama.cpp for response generation grounded in retrieved knowledge. Enabled multilingual voice interaction in English, Hindi, and Marathi through speech-to-text, translation, and text-to-speech integration. Implemented adaptive quizzes for personalized learning and optimized the application for offline-first usage in low-connectivity environments.

SecureFin: YOLO-Based Automated Video KYC and AML Fraud Detection System

Key Skills:

YOLOv8 | Tesseract OCR | DeepFace | TensorFlow | Computer Vision | Object Detection | Image Processing | Isolation Forest

Project Link: https://github.com/kompalwargangotri/securefin-video-kyc-aml

Developed an AI-powered Video KYC and Anti-Money Laundering (AML) fraud detection system for automated identity verification and suspicious activity detection. Integrated Tesseract OCR to extract and validate Aadhaar/PAN details, YOLOv8 for person detection and single-person verification, and DeepFace for facial recognition and biometric identity verification. Implemented liveness detection to strengthen video-based identity checks and Isolation Forest for identifying potentially suspicious AML activity. Built a FastAPI backend to connect these components into a multi-stage verification pipeline, with SQLite for structured data management and storage.

NephroPredict: AI-Based Early Detection of Chronic Kidney Disease (CKD)

Key Skills:

Python | Machine Learning | Scikit-learn | Pandas | NumPy | Data Preprocessing | Feature Engineering | Model Evaluation | Data Visualization

Project Link: https://github.com/kompalwargangotri/nephropredict-ai-ckd-detection

Developed a machine learning-based system for early prediction of Chronic Kidney Disease (CKD) using patient healthcare data. Performed data preprocessing and feature engineering using Pandas and NumPy, and trained a Random Forest classifier using Scikit-learn to identify potential CKD risk. Evaluated model performance using accuracy and F1-score, achieving 90.96% accuracy, and implemented the trained model for real-time risk prediction.

ASSESSMENTS / CERTIFICATIONS

Fundamentals of Deep Learning

Provider: NVIDIA

Key Skills: Neural Networks | Model Training & Evaluation | Image Classification

Gained foundational knowledge of deep learning concepts and neural networks. Learned image classification and the fundamentals of training, validation, and evaluation of deep learning models.

Advanced Course on Green Skills and Artificial Intelligence

Provider: Edunet Foundation (under Skills4Future Program in collaboration with AICTE & Shell India Markets Pvt. Ltd.)

Key Skills:

Python | NumPy | Pandas | Machine Learning | Model Training & Evaluation | Deep Learning | Neural Networks | Generative AI Basics

Learned Python programming, Machine Learning, Deep Learning, Generative AI, and sustainable AI applications through hands-on model training and evaluation.

Claude 101

Provider: Anthropic

Key Skills: Prompt Engineering | LLMs | AI Tools & Workflows | Generative AI | Natural Language Processing (NLP)

Gained foundational knowledge of Large Language Models and AI-assisted workflows. Learned prompt engineering techniques and explored applications of Generative AI and NLP for automation, problem-solving, and content generation.

Java Programming Masterclass

Provider: Alison

Key Skills: Java Programming | Object-Oriented Programming (OOP)

Completed a comprehensive course covering core Java programming concepts and Object-Oriented Programming principles, including Java syntax and application development.

WEB LINKS

- Github - https://github.com/kompalwargangotri
- Personal - https://kompalwargangotri.github.io/